Week 7 — Quantum States of Light: The DV and CV Languages

EEQT30001 Quantum Technology and Engineering, Unit 2 (Quantum Optics and Quantum Sensing)

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Learning objectives (Week 7)

1. Explain why each electromagnetic mode is quantized as a harmonic oscillator, and use $[a, a^\dagger] = 1$ to compute properties of Fock, coherent, and squeezed states.
2. Draw vacuum, coherent, and squeezed states in phase space, and state the two encodings of light-based qubits (DV vs. CV).
3. Classify a light source from $g^{(2)}(0)$ (thermal 2 / coherent 1 / single photon 0) and explain why antibunching — not the photoelectric effect — proves the photon.
4. Describe how a cavity redesigns the vacuum (Jaynes–Cummings physics, the Purcell effect) and how an SNSPD detects a single photon.

Reading: Fox, ***Quantum Optics***, Ch. 5–7. Full derivations: QO_note_2, 3, 5, and 4.

Opening puzzle. Textbooks say the photoelectric effect proved that light is made of photons. It did not: a *semiclassical* theory — quantized atoms driven by a purely classical field — reproduces every feature of the photoelectric effect, including the frequency threshold. The experiment that actually certified the photon came seventy years after Einstein: photon *antibunching*, observed in 1977 [1]. This week builds the language needed to appreciate how much work it takes to prove that light arrives "one at a time", and what non-classical light is good for.

1 Quantum States of Light: The DV and CV Languages

1.1 One mode is a harmonic oscillator

Maxwell's equations in a source-free region are linear, so their solutions organize into *eigenmodes*: field patterns that oscillate at a single frequency ω, with a shape fixed by the geometry (a plane wave in free space, a standing wave in a cavity, a guided mode in a waveguide). The energy of one mode is quadratic in the field amplitudes $(\mathbf{E}, \mathbf{B})$ — formally identical to the energy $\frac{p^2}{2m} + \frac{m\omega^2 x^2}{2}$ of a mechanical oscillator. Canonical quantization therefore proceeds exactly as for a particle in a parabolic well: promote the canonical pair to operators, impose the commutator, and change variables to ladder operators (QO_note_2, Sec. 1–2). The result, for every mode,

$$\mathcal{H} = \hbar\omega\left(a^\dagger a + \tfrac{1}{2}\right), \qquad [a, a^\dagger] = 1. \tag{1.1}$$

A *photon* is one quantum of excitation of a mode — the n-th rung of that mode's energy ladder — not a small ball flying through space. **The mode comes first**; **the photon is the mode's excitation**. Everything in this unit follows from the single algebraic statement $[a, a^\dagger] = 1$.

1.2 The vacuum is not empty

The ground state $|0\rangle$ of Eq. (1.1) has energy $\hbar\omega/2$ and, more importantly, *fluctuating fields*: $\langle 0|\mathbf{E}|0\rangle = 0$ but $\langle 0|\mathbf{E}^2|0\rangle \neq 0$. These zero-point fluctuations are physical. Summing $\hbar\omega/2$ over the modes allowed between two parallel mirrors, and comparing with free space, yields a measurable attraction (Casimir, 1948; full derivation in QO_note_2, Example 4),

$$\frac{F}{A} = -\frac{\pi^2 \hbar c}{240\, d^4}. \tag{1.2}$$

The lesson to remember for Sec. 3 and for Week 10: the vacuum has *structure* — which modes exist, and how strong their zero-point fields are at a given point, depends on the boundary conditions. Structure can be engineered.

1.3 Fock states: the DV language

The eigenstates $|n\rangle$ of $a^\dagger a$ (*Fock* or number states) are maximally quantum: the field expectation value $\langle n|\mathbf{E}|n\rangle$ vanishes for every n, so a Fock state carries *no phase* — only fluctuations (QO_note_2, Sec. 2.3). Encoding information in discrete photonic degrees of freedom — photon number ($|0\rangle, |1\rangle$), polarization ($|H\rangle, |V\rangle$), or path — is called *discrete-variable* (DV) encoding. It is conceptually clean but experimentally expensive: single photons are hard to make and easy to lose.

1.4 Quadratures and phase space: the CV language

The Hermitian combinations of a and $a^\dagger$,

$$X = \frac{a + a^\dagger}{\sqrt{2}}, \qquad P = \frac{a - a^\dagger}{i\sqrt{2}}, \qquad [X, P] = i \;\;\Rightarrow\;\; \Delta X\, \Delta P \geq \tfrac{1}{2}, \tag{1.3}$$

are the dimensionless position and momentum of the mode — the *quadratures*.[1] They are continuously valued and directly measurable (by homodyne detection, Week 10), and encoding information in them is called *continuous-variable* (CV) encoding.

Phase space gives the picture. A classical oscillator is a *point* (x, p) tracing a circle as it evolves. A quantum state cannot be a point — Eq. (1.3) forbids sharp X and P simultaneously — so even the "most classical" quantum state is a fuzzy blob of area $\sim \frac{1}{2}$, rotating rigidly at ω (Fig. 1).

1.5 Coherent states: the most classical light

Which quantum state behaves like the classical field $E_0 \cos(\omega t + \phi)$ of a laser? Demanding $\langle \mathbf{E} \rangle \propto$ the classical field leads to the eigenvalue problem (QO_note_3, Sec. 2)

$$a|\alpha\rangle = \alpha|\alpha\rangle, \qquad |\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \tag{1.4}$$

with α a complex number — the dimensionless amplitude *and* phase of the wave. Key properties:

[1] The original lecture notes QO_note_2-5 use the convention $X_s = (a + a^\dagger)/2 = X/\sqrt{2}$, $Y_s = P/\sqrt{2}$, for which $[X_s, Y_s] = i/2$ and $\sigma(X_s)\sigma(Y_s) \geq 1/4$. All statements convert by a factor $\sqrt{2}$ per quadrature.

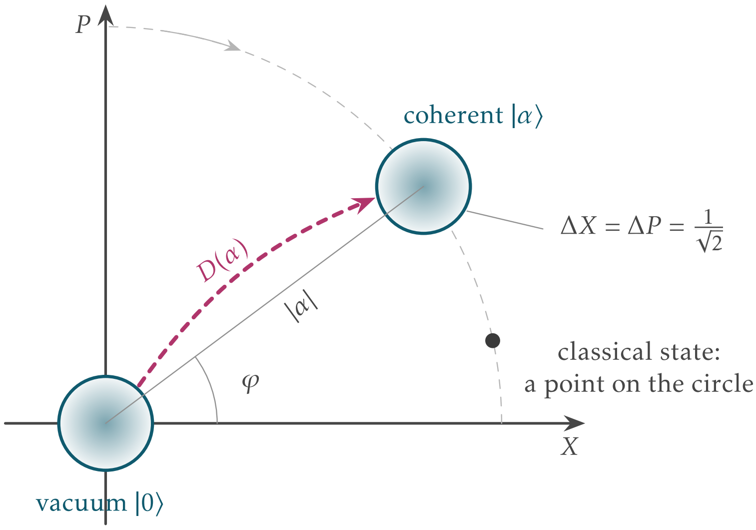

Figure 1: Phase space in our convention, Eq. (1.3). A classical state is a point tracing a circle; a quantum state is a minimum-uncertainty blob of area $\sim \frac{1}{2}$. The vacuum sits at the origin; the displacement $D(\alpha)$ shifts it to α without changing its shape — a coherent state, $\Delta X = \Delta P = 1/\sqrt{2}$.

- Photon number is Poisson-distributed, $p_n = e^{-\langle n\rangle}\langle n\rangle^n/n!$, with $\langle n\rangle = |\alpha|^2$ and $\Delta n = \sqrt{\langle n\rangle}$ — photons in a laser beam arrive ***randomly and independently***, like raindrops.
- It is a minimum-uncertainty state, $\Delta X = \Delta P = 1/\sqrt{2}$: the same circular blob as the vacuum, displaced to α. Indeed $|\alpha\rangle = D(\alpha)|0\rangle$ with the displacement operator $D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a)$ — and $D(\alpha)$ is exactly the evolution generated by a classical oscillating current (an antenna, a laser drive; QO_note_4, Sec. 3). A laser ***displaces the vacuum***.
- Free evolution just rotates the blob: $\alpha(t) = \alpha(0)e^{-i\omega t}$.

1.6 Squeezed states: reshaping the vacuum blob

The uncertainty product bounds the ***area*** of the blob, not its shape. The squeeze operator (QO_note_5, Sec. 2)

$$S(\xi) = \exp\left(\tfrac{1}{2}\xi^* a^2 - \tfrac{1}{2}\xi\,(a^\dagger)^2\right), \qquad \xi = re^{i\theta}, \tag{1.5}$$

turns the circle into an ellipse: for $\theta = 0$, $\Delta X = e^{-r}/\sqrt{2}$ and $\Delta P = e^{+r}/\sqrt{2}$ — one quadrature quieter than the vacuum itself, at the price of the other (Fig. 2). Because $S(\xi)$ is quadratic in $a^\dagger$, generating squeezed light requires a ***nonlinear*** process ($\chi^{(2)}$ parametric down-conversion, Week 8). Why care? Metrology: LIGO injects squeezed vacuum to measure below the vacuum noise level (Week 10).

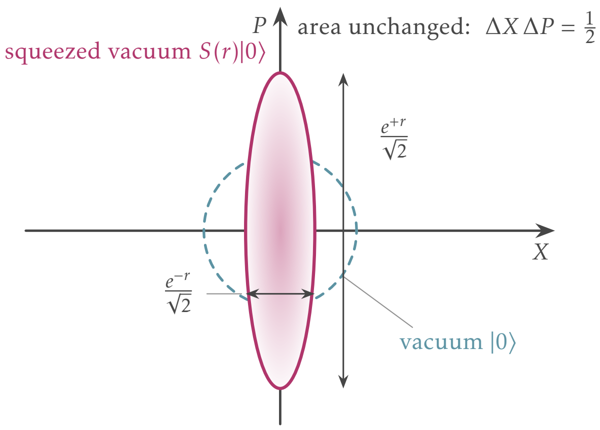

Figure 2: Squeezed vacuum in phase space: one quadrature narrows to $e^{-r}/\sqrt{2}$ at the price of the conjugate one; the area $\Delta X\,\Delta P = \frac{1}{2}$ is unchanged (dashed: the vacuum circle).

1.7 DV vs. CV at a glance

	DV (discrete)	**CV (continuous)**
Basic objects	Fock states, polarization	quadratures X, P
Qubit example	$\|H\rangle, \|V\rangle$; dual-rail $\|01\rangle, \|10\rangle$	coherent/squeezed states; cat, GKP codes
Measurement	single-photon detectors	homodyne/heterodyne
Strength	clean logic, long-distance QKD	deterministic sources, scalable
Weakness	probabilistic sources, loss	noise accumulates

Hybrid schemes (converting DV qubits into cat-state qubits and back) are an active research direction (QO_note_3, introduction). This DV/CV pair of languages is the backbone of the whole unit: Week 8 speaks mostly DV, Week 9 uses both, Week 10 mostly CV.

Summary: Section 1

Every mode of light is a harmonic oscillator: $[a, a^\dagger] = 1$, $\mathcal{H} = \hbar\omega(a^\dagger a + \frac{1}{2})$. The vacuum is a real physical state with fluctuating fields (Casimir). Fock states have sharp n and no phase (DV); coherent states are displaced vacua with Poisson statistics, $\Delta n = \sqrt{\langle n\rangle}$ (the most classical light); squeezed states beat the vacuum noise in one quadrature (CV). Phase space: point (classical) $\rightarrow$ blob of area $\sim \frac{1}{2}$ (quantum).

2 Proving the Photon: The Second-Order Correlation $g^{(2)}(0)$

2.1 Photon-counting statistics

Modern detectors do not record a continuous intensity; they *click*. The statistics of clicks classify light (QO_photon_stat):

- **Poissonian**, $\Delta n = \sqrt{\langle n\rangle}$: coherent light (laser).
- **Super-Poissonian**, $\Delta n > \sqrt{\langle n\rangle}$: thermal/chaotic light.
- **Sub-Poissonian**, $\Delta n < \sqrt{\langle n\rangle}$: *no classical explanation* — non-classical light.

A practical warning: loss randomizes. If photons survive a lossy channel or an inefficient detector independently with probability η, sub-Poissonian statistics degrade toward Poissonian — observing quantum statistics requires high collection and detection efficiency. This motivates the detectors of Sec. 3.3.

2.2 The HBT experiment and $g^{(2)}$

Photon counting alone cannot separate "one photon at a time" from a very dim laser (see the discussion question below). The decisive quantity is the ***intensity–intensity*** correlation, measured in the Hanbury Brown–Twiss (HBT) geometry: split the beam 50:50 onto two detectors and count coincidences (Fig. 3). For a single mode, with the detector-ordering of operators from Glauber's photodetection theory (QO_note_5, Sec. 1),

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a\rangle}{\langle a^\dagger a\rangle^2} = 1 + \frac{\sigma^2(n) - \langle n\rangle}{\langle n\rangle^2}. \tag{2.1}$$

$$g^{(2)}(0) = \begin{cases} 2 & \text{thermal light (bunching),} \\ 1 & \text{coherent light (random arrivals),} \\ \dfrac{n-1}{n} \xrightarrow{n=1} 0 & \text{Fock state (antibunching).} \end{cases} \tag{2.2}$$

For *any* classical intensity distribution, the Cauchy–Schwarz inequality forces $g^{(2)}(0) \geq 1$: classical waves can bunch, never antibunch. Therefore a measured $g^{(2)}(0) < 1$ has no classical explanation, and $g^{(2)}(0) \to 0$ certifies a single-photon source. This is the experiment that finally proved the photon: resonance fluorescence of single atoms, Kimble, Dagenais and Mandel (1977) [1] — seventy years after the photoelectric effect. The photoelectric effect needs only quantized *atoms*; antibunching needs quantized *light*.

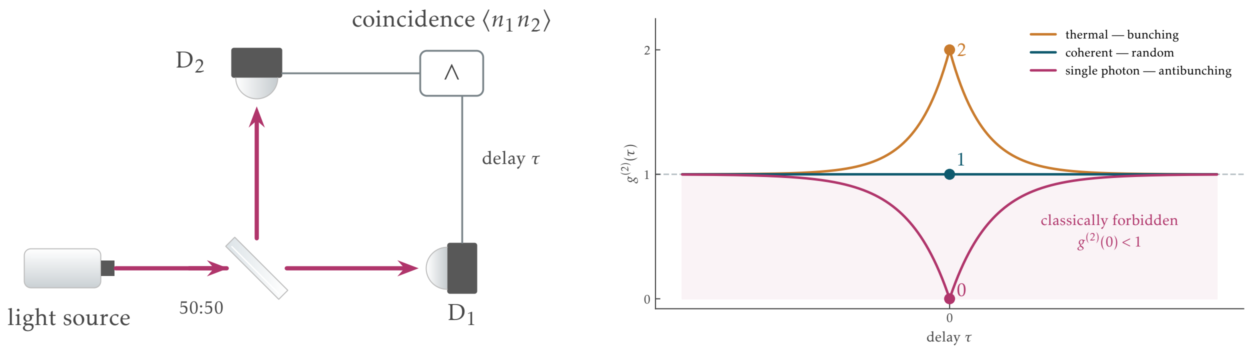

Figure 3: Left: HBT setup — beam splitter, two detectors, coincidence counter. Right: $g^{(2)}(\tau)$ for thermal, coherent, and single-photon light; the shaded region $g^{(2)}(0) < 1$ is classically forbidden.

In-class discussion 1. *A laser is attenuated to* 0.1 *photons per pulse on average. Is it a single-photon source?* — No. Attenuation preserves coherent-state statistics: $g^{(2)}(0) = 1$, and the two-photon probability $p_2 \approx \langle n\rangle^2/2$ is nonzero. Dim is not quantum.

In-class discussion 2. *You measure* $g^{(2)}(0) = 0.5$. *What is the source?* — Non-classical (below 1), but not a perfect single-photon source: e.g. two independent single-photon emitters ($g^{(2)} = 1/2$ exactly), or one imperfect emitter with background.

2.3 Try it in three lines (QuTiP)

QuTiP demo: $g^{(2)}(0)$ of Fock, coherent, and squeezed states

```
from qutip import *
N = 30
states = {"Fock |1>": basis(N,1), "Coherent": coherent(N,2.0),
          "Squeezed": squeeze(N,1.0)*basis(N,0)}
a = destroy(N)
for name, psi in states.items():
    g2 = expect(a.dag()*a.dag()*a*a, psi)/expect(a.dag()*a, psi)**2
    print(name, "g2(0) =", round(g2,3))
```

This is the seed of the bonus homework (published after this week; Colab recommended; worth up to +5 points on Midterm 2).

2.4 Single-photon sources compared

Source	$g^{(2)}(0)$ (typ.)	Rate	Operating T	Deterministic?
Attenuated laser	1 (always)	GHz	room	no (Poisson)
SPDC, heralded (Week 8)	10^{-2} (best 3×10^{-4})	0.1–4 MHz	room	heralded
Quantum dot in cavity	$\sim 10^{-3}$	tens of MHz	4–10 K	near-det.
NV center in diamond	0.2–0.4	10^5–3×10^6 s^{-1}	room	det., poor collection

Table 1: Order-of-magnitude comparison of single-photon sources; figures verified against the literature as of 2026-07 (cf. Fox Ch. 5; [3, 4]).

The quantum-dot route is where cavity QED (Sec. 3) pays off: a cavity both speeds up emission (Purcell effect) and funnels the photons into one collectible mode.

Summary: Section 2

Click statistics classify light; $g^{(2)}(0) = 1 + [\sigma^2(n) - \langle n\rangle]/\langle n\rangle^2$ is the sharp tool: thermal 2, coherent 1, single photon 0. Classically $g^{(2)}(0) \geq 1$; antibunching is the fingerprint of the photon (1977). A dim laser is never a single-photon source. Real sources trade off purity, rate, temperature, and determinism.

3 Cavity QED: Redesigning the Vacuum

3.1 The Jaynes–Cummings model in one page

Put a two-level atom ($|E_v\rangle, |E_c\rangle$, splitting $\hbar\omega_{cv}$) in a cavity supporting a single mode of frequency ω. With the rotating-wave approximation, light–matter interaction reduces to the Jaynes–Cummings Hamiltonian (QO_note_4, Sec. 4)

$$\mathcal{H}_{\mathrm{JC}} = \hbar\omega\, a^\dagger a + \frac{\hbar\omega_{cv}}{2}\sigma_z + \hbar\lambda\left(\sigma_+ a + \sigma_- a^\dagger\right), \tag{3.1}$$

where λ is the coupling rate set by the dipole moment and the mode's vacuum field. Because the excitation number is conserved, the dynamics splits into 2×2 blocks $\{|E_c\rangle|n\rangle, |E_v\rangle|n+1\rangle\}$: the atom and the cavity coherently swap one quantum at the Rabi frequency $2\lambda\sqrt{n+1}$ (on resonance). The eigenstates — *dressed states* — are split by $2\hbar\lambda\sqrt{n+1}$.

The striking case is $n = 0$: even with ***no photon in the cavity***, the excited atom oscillates, $|E_c\rangle|0\rangle \leftrightarrow |E_v\rangle|1\rangle$ — the ***vacuum Rabi oscillation*** (Fig. 4). The "empty" cavity is not empty; its zero-point field drives the atom. (With a coherent drive the oscillations collapse and revive — see QO_note_4, Sec. 4.3, for this beautiful signature of field quantization.)

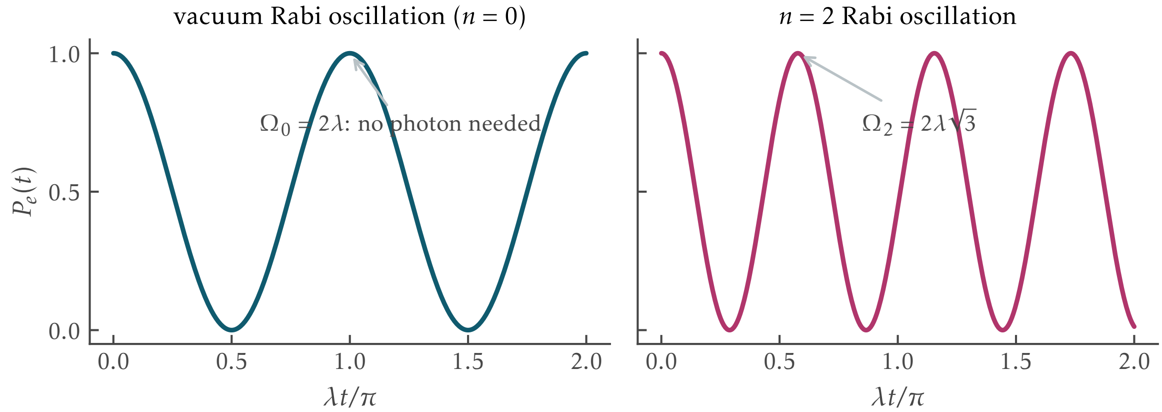

Figure 4: Jaynes–Cummings Rabi oscillations, $P_e(t) = \cos^2\!\left(\lambda\sqrt{n+1}\,t\right)$, for $n = 0$ (vacuum) and $n = 2$; the oscillation persists at $n = 0$.

3.2 Spontaneous emission and the Purcell effect

Fermi's golden rule says the emission rate of an excited atom is proportional to the density of final photon states at its transition frequency (QO_note_4, Sec. 2.2):

$$W = \frac{\pi}{2\hbar^2}\left|\langle c|\mathcal{H}_I|v\rangle\right|^2 \rho(\omega)\hbar \implies W_{\text{free}} = \frac{\omega^3|d_{cv}|^2}{3\pi\epsilon_0\hbar c^3}, \tag{3.2}$$

where the free-space result uses the vacuum density of modes $g(\omega) = \omega^2/\pi^2c^3$ (QO_note_2, Sec. 3.1). Read this formula the other way around: *spontaneous emission is not a property of the atom alone.* It is stimulated by the vacuum fluctuations of the modes around the atom — so if you change the modes, you change the rate. A cavity of quality factor Q and mode volume V concentrates the vacuum field at its resonance; the emission rate is enhanced by the Purcell factor (Purcell, 1946 [2]; result only)

$$F_P = \frac{W_{\text{cav}}}{W_{\text{free}}} = \frac{3}{4\pi^2}\left(\frac{\lambda_0}{n}\right)^3 \frac{Q}{V}, \tag{3.3}$$

for an emitter on resonance and at the field maximum. Detuned from the cavity, emission is *suppressed* instead. **Change the structure of the vacuum, and you change the atom's fate**. Keep Eq. (3.3) in mind: in Week 10 we will see that nanophotonics is precisely the engineering of Q/V — of the vacuum itself — and that this turns quantum-sensing limits into design parameters.

In-class discussion 3. *An atom is placed in a resonant cavity. Which of its properties change?* — Its emission rate (faster on resonance, slower off resonance), its emission direction (into the cavity mode), and even its energy levels (dressed-state splitting). The vacuum has been redesigned around it.

3.3 Detecting one photon: the SNSPD

Certifying $g^{(2)}(0) \approx 0$ requires detectors that click on a single photon. Photomultipliers (PMTs) and avalanche photodiodes (SPADs) do this with quantum efficiencies of order 10–70% and nanosecond timing. The current workhorse of quantum optics is the *superconducting nanowire single-photon detector* (SNSPD): a ~ 100-nm-wide superconducting wire (NbN, NbTiN, or WSi), cooled below its critical temperature (~1–4 K) and biased just below its critical current. One absorbed photon creates a local hotspot, the wire momentarily turns resistive, and a voltage pulse marks the arrival time. State-of-the-art SNSPDs combine system detection efficiency above 90% — the record is $98.0 \pm 0.5\%$ at 1550 nm [5] — with dark-count rates that can be pushed below 1 Hz (milli-Hz demonstrated) and timing jitter down to 2.6–4.3 ps [6]. Every experiment in Weeks 8–9 — QKD, Bell tests, boson sampling — stands on these detectors.

3.4 Coda: Glauber's photon

Roy Glauber received the 2005 Nobel Prize for the quantum theory of optical coherence — the theory behind $g^{(1)}$, $g^{(2)}$, and the coherent states used throughout this week. His central insight is the one we started from: light is described by quantum states of *modes*, and a photon is nothing more (and nothing less) than a single excitation of one of them. **Modes first**; **photons second**. Next week we take two photons at a time and let them interfere.

Summary: Section 3

JC physics: atom + cavity mode swap one quantum at rate $2\lambda\sqrt{n+1}$; vacuum Rabi oscillations exist at $n = 0$. Spontaneous emission $\propto$ density of vacuum modes $\Rightarrow$ Purcell factor $F_P = \frac{3}{4\pi^2}(\lambda_0/n)^3 Q/V$: the vacuum is designable. SNSPDs (superconducting nanowires) detect single photons with $> 90\%$ efficiency and ps jitter, enabling all the quantum optics that follows.

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